Practical No. 14

File Upload and Download Automation

Aim: Automate file upload and download, validate successful upload, and verify that a downloaded file exists.

Steps

1. Ensure JDK, Eclipse, Chrome, and Selenium Java libraries are installed.
2. Create folders **D:\STQA\FileUploadDownload\upload** and **D:\STQA\FileUploadDownload\download**.
3. Create **student.txt** in the upload folder and add: *This is a Selenium file upload test. Student Record*.
4. Create a Java project named **FileUploadDownload**, then package **Fileupload** and class **FileUploadDownload**.
5. Add the Selenium JAR files to the project's Build Path.
6. Paste the Java code, check the file paths, save, and run as **Java Application**.

Code

FileUploadDownload.java

```
package Fileupload;
import java.awt.Robot;
import java.awt.Toolkit;
import java.awt.datatransfer.StringSelection;
import java.awt.event.KeyEvent;
import java.io.File;
import java.time.Duration;
import java.util.HashMap;
import java.util.Map;
import org.openqa.selenium.By;
import org.openqa.selenium.JavascriptExecutor;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
public class FileUploadDownload {
    public static void main(String[] args) throws
Exception {
        // FILE PATHS
        String uploadFilePath =

"D:\\STQA\\FileUploadDownload\\upload\\student.txt";
        String downloadFolderPath =
                "D:\\STQA\\FileUploadDownload\\download";
        // CHECK UPLOAD FILE
        File uploadFile =

                new File(uploadFilePath);
        if (!uploadFile.exists()) {
            System.out.println(
                    "ERROR: Upload file not found!");
            System.out.println(
                    "File path: " + uploadFilePath);
            return;
        }
        System.out.println(
                "Upload file found:");
        System.out.println(
                uploadFilePath);
        // CREATE DOWNLOAD FOLDER
        File downloadFolder =
                new File(downloadFolderPath);
        if (!downloadFolder.exists()) {
            downloadFolder.mkdirs();
            System.out.println(
                    "Download folder created.");
        }
        // CHROME OPTIONS
        ChromeOptions options =
                new ChromeOptions();
        Map<String, Object> prefs =
                new HashMap<>();
        prefs.put(
                "download.default_directory",
                downloadFolderPath);
        prefs.put(
                "download.prompt_for_download",
                false);
```

```
prefs.put(
        "download.directory_upgrade",
        true);
options.setExperimentalOption(
        "prefs",
        prefs);
// START CHROME
WebDriver driver =
        new ChromeDriver(options);
driver.manage()
        .window()

        .maximize();
try {
    // FILE UPLOAD
    System.out.println();
    System.out.println(
            "Opening File Upload page...");
    driver.get(
            "https://the-internet.herokuapp.com/upload");
    Thread.sleep(2000);
    // FIND CHOOSE FILE ELEMENT
    WebElement uploadElement =
            driver.findElement(
                    By.id("file-upload"));
    // OPEN WINDOWS FILE DIALOG
    JavascriptExecutor js =
            (JavascriptExecutor) driver;
    js.executeScript(
            "arguments[0].click();",
            uploadElement);
    Thread.sleep(2000);
    // ROBOT CLASS
    Robot robot =
            new Robot();
    robot.setAutoDelay(100);
    // COPY FILE PATH
    StringSelection selection =
            new StringSelection(
                    uploadFilePath);
    Toolkit.getDefaultToolkit()
            .getSystemClipboard()
            .setContents(
                    selection,
                    null);

    Thread.sleep(1000);
    // PASTE FILE PATH
    robot.keyPress(
            KeyEvent.VK_CONTROL);
    robot.keyPress(
            KeyEvent.VK_V);
    robot.keyRelease(

            KeyEvent.VK_V);
    robot.keyRelease(
            KeyEvent.VK_CONTROL);
    Thread.sleep(1000);
    // PRESS ENTER
    robot.keyPress(
            KeyEvent.VK_ENTER);
    robot.keyRelease(
            KeyEvent.VK_ENTER);
    Thread.sleep(3000);
    System.out.println(
            "File selected using Robot Class.");
    // CLICK UPLOAD
    System.out.println();

    System.out.println(
            "Clicking Upload button...");
    driver.findElement(
            By.id("file-submit"))
            .click();
    // WAIT FOR UPLOAD RESULT
    WebDriverWait wait =
            new WebDriverWait(
                    driver,
                    Duration.ofSeconds(10));
    wait.until(
            ExpectedConditions
                    .presenceOfElementLocated(
                            By.tagName("body")));
    Thread.sleep(2000);
    // GET UPLOAD RESULT
    String uploadResult =
            driver.findElement(
                    By.tagName("body"))
                    .getText();
    System.out.println();
    System.out.println(
            "Upload Result:");
    System.out.println(
```

```
                uploadResult);
            // VERIFY UPLOAD
            if (uploadResult.contains(

                    "File Uploaded!")) {
                System.out.println();
                System.out.println(
                        "File Upload Successful");
            } else if (uploadResult.contains(
                    "Internal Server Error")) {
                System.out.println();
                System.out.println(
                        "File Upload Failed");
                System.out.println(
                        "Server returned Internal Server Error.");
            } else {
                System.out.println();
                System.out.println(
                        "File Upload Failed");
            }
            // FILE DOWNLOAD
            System.out.println();
            System.out.println(
                    "Opening File Download page...");
            driver.get(
                    "https://the-internet.herokuapp.com/download");
            Thread.sleep(2000);
            // COUNT FILES BEFORE DOWNLOAD
            File[] filesBefore =
                    downloadFolder.listFiles();
            int filesBeforeCount =
                    filesBefore == null
                            ? 0
                            : filesBefore.length;
            System.out.println(
                    "Files before download: "
                            + filesBeforeCount);
            // CLICK DOWNLOAD LINK
            driver.findElement(
                    By.cssSelector("#content a"))
                    .click();
            System.out.println(
                    "Download started...");
            // WAIT FOR DOWNLOAD

            File downloadedFile =
                    null;
            boolean downloadSuccessful =
                    false;
            for (int i = 0; i < 30; i++) {
                Thread.sleep(1000);
                File[] files =
                        downloadFolder.listFiles();
                if (files != null) {
                    for (File file : files) {
                        // Ignore Chrome temporary files
                        if (file.getName()
                                .endsWith(".crdownload"))
{
                            continue;
                        }
                       // Check whether this is a new file
                        if (file.lastModified()
                                >
System.currentTimeMillis() - 30000) {
                            downloadedFile =
                                    file;
                            downloadSuccessful =
                                    true;
                            break;
                        }
                    }
                }
                if (downloadSuccessful) {
                    break;
                }
            }
            // VERIFY DOWNLOAD
            System.out.println();
            if (downloadSuccessful
                    && downloadedFile != null) {
                System.out.println(
                        "File Download Successful");
                System.out.println(
                        "Downloaded File: "
                                + downloadedFile

                                .getName());
                System.out.println(
                        "Downloaded File Path: "
                                + downloadedFile
                                .getAbsolutePath());
            } else {
```

```
                System.out.println(
                        "File Download Failed");
            }
        } finally {
            // CLOSE BROWSER
            driver.quit();

            System.out.println();

            System.out.println(
                    "Browser closed.");
        }
    }
}
```